



Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories to guide strategic business decisions.

Dataset at a Glance

3,900

Purchases

Demographics

Age, Gender, Location, Subscription

18

Columns

Purchase Details

Item, Category, Amount, Season, Size

37

Missing Ratings

Behavior

Discounts, Frequency, Ratings, Shipping

Data Preparation in Python

01

Load & Explore

pandas, df.info(), .describe()

02

Clean & Impute

Missing ratings filled with category median

03

Engineer Features

age_group bins, purchase_frequency_days

04

Load to PostgreSQL

Cleaned DataFrame pushed to database



SQL Analysis: Revenue Insights

Revenue by Gender

Compared total revenue — male vs. female customers.

Revenue by Age Group

Identified which age segments drive the most spend.

Shipping Type

Standard vs. Express — average purchase amount compared.



Top 5 Products by Rating

SQL query identified products with the **highest average review ratings** across all categories.

1

Rank 1

Highest avg. review rating

2

Rank 2

Strong customer satisfaction

3

Rank 3

Consistent positive feedback

4

Rank 4

Above-average ratings

5

Rank 5

Notable customer approval

Subscribers vs. Non-Subscribers



Key Finding

Subscribers show **higher average spend** and greater total revenue contribution.

Repeat Buyers

Customers with **>5 purchases** are significantly more likely to subscribe.

- ✓ Subscription status is a strong predictor of lifetime value.

Customer Segmentation



New

First-time buyers



Returning

Moderate purchase history



Loyal

High-frequency buyers

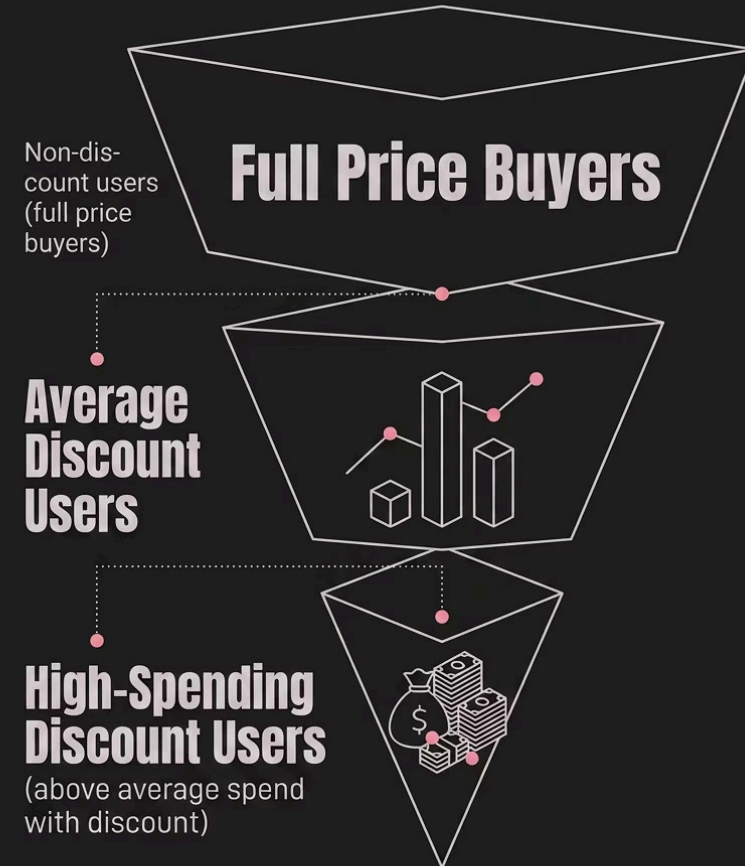
Classified via **purchase history thresholds** in SQL. Also identified **top 3 products per category** and **discount-dependent products** (5 highest % discounted).

High-Spending Discount Users

The Finding

A segment of customers used discounts yet still spent **above the average purchase amount** — high-value targets for retention.

- ❶ Discounts don't always reduce revenue — they can attract premium spenders.



Power BI Dashboard

Interactive dashboard built to visualize all key findings — revenue trends, segmentation, product performance, and subscription behavior — in one unified view.



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers → move to "Loyal"



Review Discount Policy

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users